

MAC

- Media access control
- It is a physical address *MAC*
- It is in hexa decimal format
- It contains 12 hexa decimal
- Each hexa contains 4 bit
- $12 \times 4 = 48$ bit length

IP Address

Can error in these numbers 8 bit and time

MAC

7C-F1-7E-69-49-83

Vendor

NIC

24 bit | 24 bit

Vendor ID | User ID

S. Subnet

IP

- Internet protocol
- It is a logical address
- It is in dotted decimal format
- It contains 4 octet
- Each octet contains 8 bit
- $8 \times 4 = 32$ bit length
- IP = network + host

IPv4 Structure

32 bit

Binary Combination

1st 2nd 3rd 4th

0 0

0.0.0.0 Starting IP

255.255.255.255 End IP

Class

A B C D E

Types of Communication

Unicast one → one

Multicast one → group

Broadcast one → All

CLASS	RANGE	PORTION
A	0 - 127	N.H.H.H
B	128 - 191	N.N.H.H
C	192 - 223	N.N.N.H
D	224 - 239	Multicast
E	240 - 255	Broadcast RAO

50.0.0.10 - A - N.H.H.H

192.1.255.192 - B - N.N.H.H

192.255.176.80 - C - N.N.N.H

Network - Street No

Host - Door No

How many n/w bits? 24

How many n/w octet? 3

How many host octet? 1

How many host bit? 8

SUREN School

Binary Combination

2 bit

2 1

0	0
0	1
1	0
1	1

3 bit

4 2 1

0	0	0
0	0	1
0	1	0
0	1	1
1	0	0
1	0	1
1	1	0
1	1	1

5th bit

5th bit

50 . 0 . 0 . 92

Host octet ✓
N/w bit ✓
N/w octet ✓